

Genetic report for longevity

Mood and Depression Risk

Polygenic risk of depression, mood instability, bipolar and lithium spectrum, serotonin and BDNF, and genetics of mood drugs (low dose, lithium, isotretinoin, antidepressants)

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Depression has a heritability of about 35% and is highly polygenic: hundreds of loci with very small effect, without any "depressive" gene. The classical candidate genes (5-HTTLPR, BDNF, HTR2A) were not confirmed in the large GWAS; therefore this report gives more value to polygenic scores and to the GWAS loci and reads the serotonin genes without judgment, as in the profile (§14, §15) and in the Dopamine and Noradrenaline reports. Adds the bipolar spectrum (already read in the Psychedelic report), the instability of mood and genetics of drugs in use: low dose lithium (5 mg), low dose naltrexone (4.5 mg) and isotretinoin. It does not take SSRIs or MAOIs. A genetic score measures a inherited trend, not a diagnosis or a destination; the environment, sleep, exercise and relationships count more than genes.

Only genes: genotyping chip + local imputation + polygenic scores (PGS Catalog) and loci scores of GWAS Catalog, with percentiles in the reference population. Each theme is divided into favourable and unfavourable and linked to the aging mechanisms of the list of objectives. It is neither a diagnosis nor a prescription. All practical ideas are marked as a suggestion (not applied).

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Summary

23

variants

15

in the chip

7

imputed
($DR2 \geq 0.8$)

1

not found

6

Scores with
percentile

Overall reading. The genetic risk of depression is above average in the most complete score, and this is the point of attention of the report. The genomic score of major depression (943 thousand variants) is at the 95th percentile; that of having already been depressed for an entire week in 79. The other two are typical: depression (7.5 thousand variants) in 58 and score 268 loci of GWAS in 40. In strong loci: TMEM161B C/C against; DRD2, SORCS3 and PCLO without risk alleles in favor; LRFN5 and RERE typical. In the bipolar spectrum, the genomic score of the psychedelic report is at the 79th percentile, but the 87 loci score is in the 29 and FADS2 A/A is favourable; CACNA1C, ANK3, MAD1L1 and TRANK1 are typical. Mood instability at the 66th percentile. Genetic insomnia (the 89th percentile, Circadian Rhythm and Sleep report) is the most modifiable risk factor that counts with these scores. In serotonin and BDNF, without judgment as in the profile: BDNF Val/Met, 5-HTTLPR contradictory (§15), HTR1A C/G, HTR2A rs6311 C/C, TPH2 G/T, COMT Val/Val. In drugs: GSK3B G/G (target lithium, "typical"), OPRM1 A/G (1 allele Asp40, informative for naltrexone), MTHFR 677 C/T. CYP2D6 *1/*2 and CYP2C19 *1/*1: if one day an antidepressant, normal metabolizer is needed. None of this is diagnostic; genetics counts less than sleep, exercise, relationships and drugs in use (isotretinoin).

Favourable

- **DRD2 rs61902811 AA:** DRD2: dopamine D2 receptor; locus of depression, well-being and anxiety.
- **SORCS3 rs1021363 GG:** SORCS3: neuronal receptor; shared with ADHD.
- **PCLO rs2247523 CC:** PCLO (piccolumn): active zone of synapse.
- **FADS2 rs174592 AA:** FADS2: omega-3 fatty acids desaturase and omega-6; bipolar disease locus.

Unfavourable

- **Major depression (943 784 variants):** the 95th percentile
- **It was already depressed a whole week (5777 variants):** the 79th percentile
- **Bipolar disease (PGS002786, 948996 variants):** the 79th percentile (Psicodelic report)
- **Insomnia (129 loci):** the 89th percentile (Circadian Rhythm and Sleep Report)
- **TMEM161B rs3099439 CC:** TMEM161B: locus of depression in the 5 chromosome.

Here, the scores out of the typical range (≤ 25 or ≥ 75 , in the trait direction) and the variants with strong or moderate evidence. The weak evidence variants are in the tables of each theme.

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP/Ensembl. Of the 15 chip genotypes, 0 needed complement, 0 are palindromic (A/T or C/G; orientation confirmed by the position and reference alleles of the chip, which is already in the strand +) and 0 were read by the position (merged rsID). The imputations come from the local VCF, also in the strand +. The cited effect alleles of studies that use the strand of the gene were converted to the strand +; therefore, some genotypes appear with different letters from those in the literature (e.g., genes in the strand – such as COL1A1 or VDR).

Polygenic risk of depression

Major depression scores (943 thousand variants and 7.5 thousand variants), having been depressed for an entire week, and the GWAS locus score. The strongest loci (NEGR1, DRD2, LRFN5, RERE, SORCS3) move in synapse and neuronal development.

Ageing mechanisms: Intercellular communication alteration; Neuroinflammation

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Major depression (943 784 variants)	Unfavourable	the 95th percentile · z=1.71	Higher score : higher risk. PGS002789 (PGS Catalog; SDPR; Major depressive disorder; PGP000370, PMID 36028495). Variants with genotype: 931235 of 943784 (covering of 98.7% coefficients).
Depression (7534 variants)	Typical / intermediate	the 58th percentile · z=0.24	Higher score : higher risk. PGS001829 (PGS Catalog; Penalized regression (bigstatsr); Depression; PGP000263, PMID 34995502). Variants with genotype: 7338 of 7534 (covering of 98.0% coefficients).
It was already depressed a whole week (5777 variants)	Unfavourable	the 79th percentile · z=0.76	Higher score = more probability. PGS001898 (PGS Catalog; Penalized regression (bigstatsr); Ever depressed for a whole week; PGP000263, PMID 34995502). Variants with genotype: 5641 of 5777 (covering of 97.3% coefficients).
Depression (Loci of GWAS)	Typical / intermediate	the 40th percentile · z=-0.28	Higher score : higher risk. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 268 of 270 loci with genotype. Studies: Als TD (Depression); GENDEP Investigators (Major depressive disorder); Giannakopoulou O (Depression); Guindo-Martinez M (Major depressive disorder); Hall LS (Major depressive disorder); Howard DM (Depression); Huider F (Major depressive disorder); Nagel M (Depression). Variants with genotype: 268 of 270 (covering of 99.3% coefficients).

Favourable

- DRD2 rs61902811 AA
- SORCS3 rs1021363 GG
- PCLO rs2247523 CC

Unfavourable

- Major depression (943 784 variants) (the 95th percentile)
- One week (5777 variants) (the 79th percentile) has been depressed
- TMEM161B rs3099439 CC

Favourable variants (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
DRD2	rs61902811 11:113500036 · G/A/C	AA	imputation DR2=1.00	G (0; 64%)	moderate	DRD2: dopamine D2 receptor; locus of depression, well-being and anxiety. Meaning: GWAS Catalog: Depression (Howard DM 2019; p=4e-39); G ↑ allele
SORCS3	rs1021363 10:104851081 · A/C/G/T	GG	imputation DR2=0.99	A (0; 34%)	moderate	SORCS3: neuronal receptor; shared with ADHD. Meaning: GWAS Catalog: Depression (Als TD 2023; p=2e-25); Allele A ↑
PCLO	rs2247523 7:82825088 · C/G/T	CC	imputation DR2=1.00	G (0; 47%)	moderate	PCLO (piccolumn): active zone of synapse. Meaning: GWAS Catalog: Depression (Howard DM 2019; p=9e-14); G ↑ allele

Unfavourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
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TMEM161B	rs3099439 5:88249501 · C/T	CC	imputation DR2=0.98	C (2; 49%)	moderate	TMEM161B: locus of depression in the 5 chromosome. Meaning: GWAS Catalog: Depression (Howard DM 2019; p=2e-21); allele C ↑
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Typical / intermediate variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
LRFN5	rs61990288 14:41605523 · G/A/T	AG	imputation DR2=1.00	G (1; 49%)	moderate	LRFN5: organization of synapse. Meaning: GWAS Catalog: Depression (Howard DM 2019; p=2e-31); G ↑ allele
RERE	rs301799 1:8429242 · C/A/G/T	TC	imputation DR2=0.99	C (1; 47%)	moderate	RERE: regulation of chromatin in neuronal development. Meaning: GWAS Catalog: Depression (Howard DM 2019; p=4e-31); allele C ↑

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
NEGR1	rs2821235 — · —	—	SNP not found in the data	—	moderate	NEGR1: neuronal adhesion; one of the strongest loci of depression. No major variant of GWAS Catalog for this gene with usable genotype and sense.

Instability of mood and bipolar spectrum

Mood instability, irritability and mood oscillations (UK Biobank), bipolar disease loci (CACNA1C, ANK3, FADS2, MAD1L1, TRANK1). The bipolar genomic score comes from the Psychedelic report.

Ageing mechanisms: Intercellular communication alteration; Circadian disorders

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Instability and mood oscillations	Typical / intermediate	the 66th percentile · z=0.36	Higher score : more instability. Loci score of GWAS Catalog ($p \leq 510-8$; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 74 of 75 loci with genotype. Studies: Nagel M (Experiencing mood swings); Nagel M (Irritable mood); Ward J (Mood instability). Variants with genotype: 74 of 75 (covering of 98.7% coefficients).
Bipolar disease (GWAS loci)	Typical / intermediate	the 29th percentile · z=-0.62	Higher score : higher risk. Loci score of GWAS Catalog ($p \leq 510-8$; 1 variant by 500 kb window; ln(OR) or beta coefficients): 87 of 93 loci with genotype. Studies: Baum AE (Bipolar disorder); Bigdeli TB (Bipolar disorder); Choi MJ (Bipolar disorder); Hou L (Bipolar disorder); Ikeda M (Bipolar disorder); Li HJ (Bipolar disorder); Muhleisen TW (Bipolar disorder); Mullins N (Bipolar I disorder). Variants with genotype: 87 of 93 (covering of 95.2% coefficients).
Bipolar disease (PGS002786, 948996 variants)	Unfavourable	the 79th percentile	It labels "more likely". Source: Psychedelic report.
Insomnia (129 loci)	Unfavourable	the 89th percentile	Insomnia is one of the strongest risk factors for depression. Source: Circadian Rhythm and Sleep Report.

Favourable

- FADS2 rs174592 AA

Unfavourable

- Bipolar disease (PGS002786, 948 996 variants) (the 79th percentile)
- Insomnia (129 loci) (the 89th percentile)

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
FADS2	rs174592 11:61851136 · A/G	AA	imputation DR2=0.99	G (0; 39%)	moderate	FADS2: omega-3 fatty acids desaturase and omega-6; bipolar disease locus. Meaning: GWAS Catalog: Bipolar disorder (Mullins N 2021; $p=1e-13$); Allele A ↓

Typical / intermediate variants (4)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CACNA1C	rs1006737 12:2236129 · G/A	GA	chip	A (1; 32%)	strong	CACNA1C (L-type calcium channel): the classic locus of bipolar disorder. Meaning: GWAS Catalog: Bipolar disorder (Ferreira MA 2008; $p=7e-08$); Allele A ↑
ANK3	rs10994336 10:60420054 · C/T	CC	chip	T (0; 7%)	moderate	ANK3 (ankyrin G): initial segment of the axon; locus of bipolar disease. Meaning: GWAS Catalog: Bipolar disorder (Ferreira MA 2008; $p=9e-09$); allele T ↑
MAD1L1	rs4236274 7:1856777 · A/G	AG	chip	G (1; 39%)	moderate	MAD1L1: locus shared between bipolar disease and schizophrenia. Meaning: GWAS Catalog: Bipolar disorder (Hou L 2016; $p=8e-12$); G ↑ allele

TRANK1	rs6550435 3:36822998 · T/G	TG	chip	G (1; 37%)	moderate	TRANK1: bipolar disease locus; the expression rises with valproate. Meaning: GWAS Catalog: Bipolar disorder (Muhleisen TW 2014; p=2e-08); G ↑ allele
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Serotonin, BDNF and plasticity

Serotonin candidate genes (SLC6A4, HTR1A, HTR2A, TPH2), BDNF and COMT. They are left without good/bad judgment, as in the profile (§14, §15, §21): the effects are small and were not confirmed in large GWAS.

Ageing mechanisms: Intercellular communication alteration; Neuroinflammation

Favourable

None.

Unfavourable

None.

Non-judgmental variants (informative) (7)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
BDNF	rs6265 11:27658369 · C/T	CT	chip	—	moderate	BDNF Val66Met: the Met allele reduces the activity-dependent release of BDNF. It is not a strong sign in the GWAS of depression; without judgment, as in §15. C/T = Val/Met (1 copy of Met).
SLC6A4	rs2129785 17:30263512 · T/C/G	TT	chip	—	weak	SLC6A4 (serotonin carrier): short/long 5-HTTLPR marker. In §15 the two markers contradict each other; 5-HTTLPR is not read by chip. Contradicting rs11867581 (§15).
SLC6A4	rs11867581 17:30295210 · A/G/T	GG	chip	—	weak	SLC6A4: second marker of 5-HTTLPR (points to the long allele).
HTR1A	rs6295 5:63962738 · C/A/G	CG	chip	—	weak	HTR1A -1019C>G: G allele was associated in small studies with more depression and less response to SSRIs.
HTR2A	rs6311 13:46897343 · C/A/T	CC	chip	—	weak	HTR2A -1438G>A: promoter of the 5-HT2A receptor (the target of psychedelics).
TPH2	rs4570625 12:71938143 · G/A/T	GT	chip	—	weak	TPH2 -703G>T: synthesis of serotonin in the brain; inconsistent effects.
COMT	rs4680 22:19963748 · G/A	GG	chip	—	weak	COMT Val158Met Val/Val eliminates prefrontal dopamine faster ("warreiro": more stress tolerance, working memory more dependent on dopamine). G/G = Val/Val.

Mood drugs: lithium, naltrexone, antidepressants

GSK3B (target of lithium, which also activates autophagy), OPRM1 (naltrexone receptor), HTR2A rs7997012 (response to citalopram), MTHFR (folate). The CYP2D6 *1/*2 and the CYP2C19 *1/*1 (normal metabolizers) come from the §21 profile and from the Psychedelic report.

Ageing mechanisms: Intercellular communication alteration; Autophagy / Deregulated Macroautophagy

Favourable

None.

Unfavourable

None.

Typical / intermediate variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
MTHFR	rs1801133 1:11796321 · G/A/C	GA	chip	A (1; 36%)	weak	MTHFR C677T (T in the gene = A in the strand read): less activity and more homocysteine; small association with depression. Based on the use of methylfolate as an adjuvant. G/A = 677 C/T heterozygous (coincidence with the profile).

Non-judgmental variants (informative) (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
GSK3B	rs334558 3:120094435 · A/C/G/T	GG	chip	—	weak	GSK3B -50T/C: promoter of the main lithium target; T allele was associated with better lithium response in small studies. Gene in reverse strand: G/G in the strand read = C/C in the gene (without the T allele); "type" in §15.
OPRM1	rs1799971 6:154039662 · A/G	AG	chip	—	moderate	OPRM1 A118G (Asn40Asp): mu opioid receptor, target of naltrexone. The G allele was associated with the best response to naltrexone in alcohol, with mixed results; there is no data for the low dose. A/G = Asn40/Asp40 (1 copy of G allele).
HTR2A	rs7997012 13:46837850 · A/C/G/T	GG	chip	—	weak	HTR2A intron 2: allele A was associated with the best response to citalopram (STAR*D study); it did not replicate well. G/G: without the A allele connected to the best response to citalopram.

Mechanisms of ageing

Linking the results to the aging mechanisms of the list of objectives for immortality. Sorted by the number of unfavourable results.

Mechanism	Unfavourable	Favourable	Typical
Intercellular Communication	Major depression (943 784 variants) (the 95th percentile); It was already depressed one week (5777 variants) (the 79th percentile); TMEM161B CC; Bipolar disease (PGS002786, 948 996 variants) (the 79th percentile); Insomnia (129 loci) (the 89th percentile)	DRD2 AA; SORCS3 GG; PCLO CC; FADS2 AA	Depression (7534 variants) (the 58th percentile); Depression (loci of GWAS) (percentile 40); LRFN5 AG; RERE TC; Instability and mood oscillations (the 66th percentile); Bipolar disease (GWAS loci) (the 29th percentile); CACNA1C GA; ANK3 CC
Neuroinflammation	Major depression (943 784 variants) (the 95th percentile); It was already depressed one week (5777 variants) (the 79th percentile); TMEM161B CC	DRD2 AA; SORCS3 GG; PCLO CC	Depression (7534 variants) (the 58th percentile); Depression (loci of GWAS) (percentile 40); LRFN5 AG; RERE TC
Circadian disorders	Bipolar disease (PGS002786, 948 996 variants) (the 79th percentile); Insomnia (129 loci) (the 89th percentile)	FADS2 AA	Instability and mood oscillations (the 66th percentile); Bipolar disease (GWAS loci) (the 29th percentile); CACNA1C GA; ANK3 CC; MAD1L1 AG; TRANK1 TG
Autophagy / Deregulated Macroautophagy	—	—	—

Practical part

Ideas for mood, crossed with the genetic plan v2 and current medication (isotretinoin, lithium 5 mg, low dose naltrexone 4.5 mg, retatrutide, dapagliflozin, telmisartan, tadalafil, rapamycin; without SSRIs or MAOIs). The current stack remains as it is; everything is suggestion (not applied). With a depression score at the 95th percentile, the objective is to protect what most protects mood and recognize early any change, especially during isotretinoin.

1. Monitoring mood during isotretinoin

Genetics: Major p95 Depression; p89 insomnia

Isotretinoin has low mood and irritability reports (the causal link is discussed). A short record of mood and sleep (e.g. PHQ-9 once a month) makes visible a change. Low mood more than 2 weeks, new irritability or negative thoughts: talking to the doctor.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

2. Aerobic exercise and strength, every week

Genetics: Major p95 Depression

150–300 minutes of aerobic and 2–3 sessions of strength per week; in mild to moderate depression the effect is close to that of antidepressants.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

3. Regular sleep as first line

Genetics: Insomnia p89; instability of mood p66

Fixed time, daylight in the morning, dark and fresh room; melatonin 0.3–1 mg (the dose of the plan). The low dose naltrexone at night can give vivid dreams or insomnia; if it happens, talk to the doctor about taking it in the morning.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

4. Omega-3 with sufficient EPA

Genetics: FADS2 A/A; Major Depression p95

Maintain the 3.2–4 g/day of EPA+DHA of the plan, ensuring at least 1 g of EPA (the dose of the trials in mood).

V2 genetic plan and medication: In the plan (dose). • *suggestion (not applied)*

5. Lithium 5 mg stable

Genetics: GSK3B G/G; bipolar p79 (PGS)

Keep the dose regular. Telmisartan and NSAID rise lithium and low dapagliflozin; pause on days of vomiting or dehydration.

V2 genetic plan and medication: Already in the plan. • *suggestion (not applied)*

6. Methylfolate and creatine

Genetics: MTHFR 677 C/T

5-MTHF 800 µg and creatine are already in the plan; both have small trials as adjuvants in mood.

V2 genetic plan and medication: Already in the plan. • *suggestion (not applied)*

Avoiding or with caution

- Classical psychedelics (psilocybin, LSD, DMT) with lithium: reports of seizures (see Psychedelic report). *suggestion (not applied)*
- St John's wort: induces CYP3A4 (low to rapamycin) and adds serotonergic effect. *suggestion (not applied)*
- SAM-e, ashwagandha and kava: in conflict with the plan (liver; SAM-e can also precipitate hypomania with a high bipolar score). *suggestion (not applied)*
- Opioids without warning the physician that naltrexone (blocks analgesia; warns before surgeries). *suggestion (not applied)*
- Nights without sleep: sleep deprivation destabilizes mood. *suggestion (not applied)*

Limitations

- Genotyping chip + local imputation (public reference panel); imputed with $DR2 < 0.8$ treated as not found. 5-HTTLPR and other VNTR do not read by chip.
- The depression scores explain few percent of the variation; a high percentile is not diagnostic and a low percentile does not protect.
- Serotonin candidate genes remain unbiased, as in the profile (§14, §15, §21).
- There are no validated genetic markers for the effects of isotretinoin on mood or low dose naltrexone.
- Percentages against 633 people from the reference population.

Methods and sources

Genotypes: first the genotyping chip (GRCh38, strand +), by rsID and, for cast rsIDs, by the position; then the local imputation (Beagle 5.5 with a public reference panel of complete genomes), only with $DR2 \geq 0.8$. The variants that do not pass are "SNP not found in the data". The raw DNA was processed locally; to the public databases only rsIDs were sent.

Variation classification: The number of copies of the unfavourable allele is counted and compared to what was expected in the reference population: at least 0.5 copies above that expected = unfavourable; at least 0.5 below = favourable; the rest = typical/intermediate. For rare variants of large effect, having 0 copies = favourable and having ≥ 1 = unfavourable. The unfavourable alleles come from the literature (cured) or, when indicated, from the strongest association in GWAS Catalog.

Main variants per gene: for genes labeled as "GWAS Catalog", the variant with the strongest genomic association ($p \leq 5 \times 10^{-8}$) is used, mapped to this gene and with available genotype.

Polygenic scores (PGS Catalog): Harmonized coefficients for GRCh38 applied to the imputed genome (chip + $DR2 \geq 0.8$, dosage). Empirical percentile in the reference population ($n=633$), calculated with the same variants. List: PGS002789 (PGS Catalog; SDPR; Major depressive disorder; PGP000370, PMID 36028495 · PGS001829 (PGS Catalog; Penalized regression (bigstatsr); Depression; PGP000263, PMID 34995502 · PGS001898 (PGS Catalog; Penalized regression (bigstatsr); Ever depressed for a whole week; PGP000263, PMID 34995502).

GWAS Catalog loci scores: for traits without PGS, weighted sum of risk variants ($p \leq 5 \times 10^{-8}$, one per window of 500kb), with a percentile calculated in the same way.

Percentiles: ≤ 25 or ≥ 75 (in the direction of the trait) = favourable or unfavourable; between 25 and 75 = typical.

Main sources

- Howard DM et al. Genome-wide meta-analysis of depression identifies 102 independent variants and highlights the importance of the prefrontal brain regions. *Nat Neurosci* 2019.
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- Privé F et al. Portability of 245 polygenic scores. *Am J Hum Genet* 2022 (PGS001829, PGS001898).
- PGS Catalog: PGS002789 and PGS002786 (publication PGP000370, PMID 36028495; method SDPR).
- GWAS Catalog and PGS Catalog (2026); Ensembl REST (public alleles and frequencies).

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